

EXERCISES ON ALGEBRA

Advanced Program

Code: MI1036

Chapter 1

Sets, Maps, and Complex Numbers

1.1. Sets and set operations

Exercise 1. Let $A = [3; 6)$, $B = (1; 5)$, $C = [2; 4]$. Determine $(A \cap B) \setminus C$, $(A \cup B) \setminus C$, $(A \setminus C) \cap (B \setminus C)$ and $(A \setminus C) \cup (B \setminus C)$.

Exercise 2. Let

$$A = \{x \in \mathbb{R} | x^2 - 4x + 3 \leq 0\}, B = \{x \in \mathbb{R} | |x - 1| \leq 1\}, C = \{x \in \mathbb{R} | x^2 - 5x + 6 \leq 0\}.$$

Compute $(A \cup B) \cap C$, $(A \cup B) \setminus C$ and $(A \cap B) \cup C$.

Exercise 3. Let $A = \{-1, 0, 1\}$, $B = \{-2, 2\}$ và $C = \{0, 1, 2, 3\}$. Find $A \times B$, $B \times C$, $(A \times B) \cap (B \times C)$, $(A \times B) \cup (B \times C)$ and $(A \times B) \setminus (B \times C)$.

Exercise 4. Let A, B, C, D be arbitrary sets. Prove that

- | | |
|---|---|
| a) $A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C)$. | e) $(A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$. |
| b) $A \cup (B \setminus A) = A \cup B$. | f) $(A \setminus B) \cap (C \setminus D) = (A \cap C) \setminus (B \cup D)$. |
| c) $(A \setminus B) \setminus C = A \setminus (B \cup C)$. | g) $(A \cup B) \times C = (A \times C) \cup (B \times C)$. |
| d) $A \setminus (A \setminus B) = A \cap B$. | h) $(A \cap B) \times C = (A \times C) \cap (B \times C)$. |

i) Is it true that $(A \cup B) \times (C \cup D) = (A \times C) \cup (B \times D)$. If not, give a counterexample.

j) If $(A \cap C) \subset (A \cap B)$ and $(A \cup C) \subset (A \cup B)$, then $C \subset B$.

1.2. Mappings

Exercise 5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2 + 4x - 5$ and $A = \{x \in \mathbb{R} \mid -3 \leq x \leq 3\}$. Find $f(A)$ and $f^{-1}(A)$.

- b) If f and g are surjective, then $g \circ f$ is surjective.
- c) If f and g are bijective, then $g \circ f$ is bijective.
- d) If f is surjective and $g \circ f$ is injective, then g is injective.
- e) Give an example to show that $g \circ f$ is injective, but g is not injective.
- f) If g is injective and $g \circ f$ is surjective, then f is surjective.
- g) Give an example to show that $g \circ f$ is surjective but f is not surjective.

1.3. Algebraic structures

Exercise 12. Determine which of the following binary operations are associative:

- a) the operation $*$ on $\mathbb{R}$ defined by: $a * b = a + b + ab$.
- b) the operation $*$ on $\mathbb{Z}$ defined by: $a * b = a - b$.
- c) the operation $*$ on $\mathbb{Z} \times \mathbb{Z}$ defined by: $(a, b) * (c, d) = (ad + bc, bd)$.

Exercise 13. Decide which of the binary operations in the preceding exercise are commutative.

Exercise 14. Determine which of the following sets are groups under the usual addition:

- a) the set of rational numbers of absolute value < 1 .
- b) the set of rational numbers with denominators equal to 1 or 2.
- c) the set of rational numbers with denominators equal to 1, 2 or 3.

Exercise 15. Consider the set $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$ with the following binary operation $*$ defined as: for $a, b \in \mathbb{Z}_5$, $a * b = (a + b) \bmod 5$ (the remainder of $(a + b)$ divided by 5). For example $2 * 4 = 1$. Show that $\mathbb{Z}_5$ is a group under this operation $*$.

Exercise 16. Which set of the following sets is a ring/field?

- a) $X = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$
- b) $Y = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$

where the addition and the multiplication are usual as follows:

$$\begin{aligned}(a + b\sqrt{2}) + (c + d\sqrt{2}) &= (a + c) + (b + d)\sqrt{2}, \\ (a + b\sqrt{2})(c + d\sqrt{2}) &= (ac + 2bd) + (ad + bc)\sqrt{2}.\end{aligned}$$

1.4. Complex numbers

Exercise 17. Find the real parts and the imaginary parts of the following complex numbers:

- a) $z = (2 + 3i)(1 - i) + \frac{3}{2 - i}$.
- b) $z = \frac{\sqrt{3} - i}{1 + i} - \frac{\sqrt{2} + i}{i}$.

c) $z = (1 + i)^2 - (1 - i)^4$.

d) $z = 1 + (1 + i) + (1 + i)^2 + \cdots + (1 + i)^{25}$.

Exercise 18. Find $x, y \in \mathbb{C}$ such that

a) $(3 - 2i)x + (5 - 7i)y = 1 - 3i$.

b) $(1 - 3i)(2x + yi) = 1 + i$.

Exercise 19. Represent the following complex numbers as the polar forms:

a) $z = 3 \left(\cos \frac{\pi}{11} - i \sin \frac{\pi}{11} \right)$.

c) $z = -2025$.

b) $z = 3 \left(-\cos \frac{\pi}{11} + i \sin \frac{\pi}{11} \right)$.

d) $z = \frac{1}{i}$.

Exercise 20. Represent the following complex numbers as the canonical forms:

a) $(1 + i\sqrt{3})^9$.

b) $\frac{(1 + i)^{21}}{(1 - i)^{13}}$.

c) $(2 + i\sqrt{12})^5(\sqrt{3} - i)^{11}$.

Exercise 21. Find all 8th roots of $1 - i\sqrt{3}$ and all 10th roots of $z = \frac{-1 + i}{1 + i\sqrt{3}}$.

Exercise 22. Suppose $(3 + 4i)^{10} = a + bi$, with $a, b \in \mathbb{R}$. Find $a^2 + b^2$.

Exercise 23. Solve the following equations in the field of complex numbers.

a) $z^2 + z + 1 = 0$.

e) $\frac{(z + i)^4}{(z - i)^4} = 1$.

i) $z^2 + |z|^2 = 0$.

b) $z^2 + 2iz - 5 = 0$.

f) $z^8(\sqrt{3} + i) = 1 - i$.

j) $\overline{z^7} = \frac{1024}{z^3}$.

c) $z^4 - 3iz^2 + 4 = 0$.

g) $\overline{z^7} = \frac{1}{z^3}$.

k) $z^5 = 16\overline{z}$.

d) $z^6 - 7z^3 - 8 = 0$.

h) $z^4 = z + \overline{z}$.

l) $z^5 + 9\overline{z} = 0$.

Exercise 24. Let $f: \mathbb{C} \rightarrow \mathbb{C}$, $f(z) = z^4 + 1$. Find $f^{-1}(\{i\})$.

Exercise 25. a) Find a such that $2 + i$ is a root of the polynomial $p(x) = x^3 - 2x^2 - 3x + a$.

b) Find a, b such that $1 + 2i$ is a root of the polynomial $p(x) = x^3 - ax^2 + bx - (2a + 2)$.

Exercise 26. Let $\varepsilon = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$. Show that

(a) $\varepsilon + \varepsilon^2 + \varepsilon^3 + \varepsilon^4 + \varepsilon^5 + \varepsilon^6 = -1$.

(b) $\varepsilon + \varepsilon^2 - \varepsilon^3 + \varepsilon^4 - \varepsilon^5 - \varepsilon^6 = i\sqrt{7}$.

Exercise 27. Let $\epsilon = \cos \frac{2\pi}{15} + i \sin \frac{2\pi}{15}$. Show that $\varepsilon + \varepsilon^2 + \varepsilon^4 + \varepsilon^7 + \varepsilon^8 + \varepsilon^{11} + \varepsilon^{13} + \varepsilon^{14} = 1$.

Chapter 2

Matrices, System of Linear Equations

2.1-2.2. Matrix operations

Exercise 28. Let $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -1 \\ 0 & 3 & -2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 & 1 \\ -2 & 3 & 0 \\ 1 & 2 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & 2 & 1 \\ 3 & 4 & 1 \\ 2 & 0 & 2 \end{bmatrix}$.

Compute $A + BC$, $A^T B - C$, $A(BC)$, $(A + 3B)(B - C)$.

Exercise 29. Let $A = \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}$.

a) Compute $F = A^2 - 3A$.

b) Find the matrix X satisfying $(A^2 + 5I)X = B^T(3A - A^2)$.

Exercise 30. Find the matrix X such that

a) $\begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} + 2X = \begin{bmatrix} 1 & -2 \\ 5 & 7 \end{bmatrix}$.

b) $\frac{1}{2}X - \begin{bmatrix} 1 & -3 & 2 \\ 3 & -4 & 1 \\ 2 & -5 & 3 \end{bmatrix} \begin{bmatrix} 2 & 5 & 6 \\ 1 & 2 & 5 \\ 1 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 0 & -6 & 6 \\ -2 & -9 & 2 \\ -4 & -8 & 6 \end{bmatrix}$.

Exercise 31. Find a real 2×2 matrix $A \neq O$ such that a) $A^2 = O$. b) $A^2 = -I_2$.

Exercise 32. Find two real 2×2 matrices A and B such that $(A + B)^2 \neq A^2 + 2AB + B^2$.

Exercise 33. Use the following given definition to find $f(A)$:

“If $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_kx^k$, then for an $n \times n$ matrix A , $f(A)$ is defined by $f(A) = a_0I_n + a_1A + a_2A^2 + \cdots + a_kA^k$.”

a) $f(x) = x^2 - 5x + 2$, $A = \begin{bmatrix} 1 & 2 \\ 2 & -3 \end{bmatrix}$. b) $f(x) = x^2 - 7x + 6$, $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$.

c) $f(x) = x^3 - 2x^2 + 5x - 10$, $A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 2 \\ -1 & 1 & 3 \end{bmatrix}$.

Exercise 34. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Prove that $A^2 - (a + d)A + (ad - bc)I_2 = O$.

Exercise 35. Let $A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$. Check that $A^2 - 4A - 5I_2 = O$ and compute A^n ($n \in \mathbb{N}$).

Exercise 36. Compute A^n , where a) $A = \begin{bmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{bmatrix}$. b) $A = \begin{bmatrix} a & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & a \end{bmatrix}$.

Exercise 37. Let A be a square matrix. Show that

- a) AA^T , $A^T A$ and $A + A^T$ are symmetric matrices.
- b) $A - A^T$ is a skew-symmetric matrix.

Exercise 38. a) Let $A \in M_n(\mathbb{R})$ be a matrix such that $AA^T = O$. Show that $A = O$.

- b) Let $A, B \in M_n(\mathbb{R})$ be two matrices such that $AA^T + BB^T = AB^T + BA^T$. Show that $A = B$.

2.3. Linear systems of equations

Exercise 39. Solve the following systems of linear equations:

$$\text{a) } \begin{cases} 2x_1 - 5x_2 + 8x_3 = 0 \\ -2x_1 - 7x_2 + x_3 = 0 \\ 4x_1 + 2x_2 + 7x_3 = 0 \end{cases}$$

$$\text{d) } \begin{cases} 3x_1 - 5x_2 + 2x_3 + 4x_4 = 2 \\ 7x_1 - 4x_2 + x_3 + 3x_4 = 5 \\ 5x_1 + 7x_2 - 4x_3 - 6x_4 = 3. \end{cases}$$

$$\text{b) } \begin{cases} -3x_1 + 5x_2 - 7x_3 = 0 \\ -6x_1 + 7x_2 + x_3 = 0 \end{cases}$$

$$\text{e) } \begin{cases} 3x_1 - x_2 + 3x_3 = 1 \\ -4x_1 + 2x_2 + x_3 = 3 \\ -2x_1 + x_2 + 4x_3 = 4 \\ 10x_1 - 5x_2 - 6x_3 = -10. \end{cases}$$

$$\text{c) } \begin{cases} x_1 - 2x_2 + x_3 = 4 \\ 2x_1 + x_2 - x_3 = 0 \\ -x_1 + x_2 + x_3 = -1. \end{cases}$$

$$\text{f) } \begin{cases} x_1 + 2x_2 + 3x_3 + 4x_4 = -4 \\ 3x_1 + 7x_2 + 10x_3 + 11x_4 = -11 \\ x_1 + 2x_2 + 4x_3 + 2x_4 = -3 \\ x_1 + 2x_2 + 2x_3 + 7x_4 = -6 \end{cases}$$

Exercise 40. Find m such that the system $\begin{cases} x + 2y - 3z = 4 \\ 3x - y + 5z = 2 \\ 4x + y + (m^2 - 14)z = m + 2. \end{cases}$ has no solutions/exactly one solution/infininitely many solutions.

2.4. Determinant and inverse of a matrix

Exercise 41. Compute the following determinants:

$$\text{a) } A = \begin{vmatrix} 1 & 0 & 2 & -1 \\ 3 & 0 & 0 & 5 \\ 2 & 1 & 4 & -3 \\ 1 & 0 & 5 & 0 \end{vmatrix}.$$

$$\text{c) } C = \begin{vmatrix} 1 & 1 & 2 & 3 \\ 1 & 2-x^2 & 2 & 3 \\ 2 & 3 & 1 & 5 \\ 2 & 3 & 1 & 9-x^2 \end{vmatrix}.$$

$$\text{b) } B = \begin{vmatrix} a & b & c & d \\ b & a & d & c \\ c & d & a & b \\ d & c & b & a \end{vmatrix}.$$

$$\text{d) } D = \begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+z & 1 \\ 1 & 1 & 1 & 1-z \end{vmatrix}.$$

Exercise 42. Find the inverses of the following matrices:

$$\text{a) } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

$$\text{c) } C = \begin{bmatrix} 3 & -4 & 5 \\ 2 & -3 & 1 \\ 3 & -5 & 1 \end{bmatrix}.$$

$$\text{b) } B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 9 & 8 & 7 \end{bmatrix}.$$

$$\text{d) } D = \begin{bmatrix} 1 & -a & 0 & 0 \\ 0 & 1 & -a & 0 \\ 0 & 0 & 1 & -a \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Exercise 43. Find m such that $A = \begin{bmatrix} a+1 & -1 & m \\ 3 & m+1 & 3 \\ m-1 & 0 & m-1 \end{bmatrix}$ is invertible.

Exercise 44. Let $A = \begin{bmatrix} -1 & 2 & 1 \\ 2 & 3 & 4 \\ 3 & 1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 2 \\ 3 & 4 \\ 0 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 12 & 10 \\ 6 & 16 & 7 \end{bmatrix}$. Determine X fulfilling $AX + B = C^T$.

Exercise 45. Prove that if A is a skew-symmetric (or antisymmetric) matrix of order n , where n is odd, then $\det(A) = 0$.

Exercise 46. a) Let A be a square matrix of order 2025. Prove that

$$\det(A - A^T)^{2025} = 2025(\det A - \det A^T).$$

b) Let A, B be square matrices of order 2025 satisfying $AB + B^T A^T = O$. Prove that $\det A = 0$ or $\det B = 0$.

Exercise 47. Let $A, B \in M_n(\mathbb{R})$. Suppose that $AB = BA$. Show that

a) $\det(A^2 + B^2) \geq 0$.

b) $\det(A^2 + AB + B^2) \geq 0$.

Exercise 48. Let $A, B \in M_n(\mathbb{R})$. Suppose that $A^2 + B^2 = O$ and $AB - BA$ is invertible. Show that n is even.

Exercise 49. Prove that if A is a real square matrix satisfying $A^3 = A + I$, then $\det A > 0$. (Hint: $A^5 = A^2 + A + I$.)

Exercise 50. Let A, B be square matrices of the same order satisfying $AB = A + B$. Prove that $AB = BA$.

Exercise 51. Let A, B be two 3×3 matrices such that $A^2 = AB + BA$. Prove that $\det(AB - BA) = 0$. (Hint: $AB - BA = A^2 - 2BA = A(A - 2B)$).

Chapter 3

Vector spaces, rank and inverse of a matrix

3.1. Vector spaces and subspaces

Exercise 52. Determine whether V is a vector space or not in the following cases:

a) $V = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ and the operations are defined by

$$(x, y, z) + (x', y', z') = (x + x', y + y', z + z'); \quad k(x, y, z) = (|k|x, |k|y, |k|z), \quad k \in \mathbb{R}.$$

b) $V = \{x = (x_1, x_2) \mid x_1 > 0, x_2 > 0\} \subset \mathbb{R}^2$ and the operations are defined by

$$(x_1, x_2) + (y_1, y_2) = (x_1 y_1, x_2 y_2); \quad k(x_1, x_2) = (x_1^k, x_2^k), \quad k \in \mathbb{R}.$$

Exercise 53. For each of the following subsets of $\mathbb{R}^3$, determine which one is a subspace of $\mathbb{R}^3$.

a) $\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + 2x_2 + x_3 = 0\}$. d) $\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 x_2 x_3 = 0\}$.

b) $\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + 2x_2 + x_3 = 1\}$.

c) $\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + 2x_2 + x_3 > 0\}$. e) $\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 = 3x_3\}$.

Exercise 54. Let V_1, V_2 be subspaces of V and $V_1 + V_2 := \{x_1 + x_2 \mid x_1 \in V_1, x_2 \in V_2\}$. Prove that

- a) $V_1 \cap V_2$ is a subspace of V . b) $V_1 + V_2$ is a subspace of V .

Exercise 55. Let V_1, V_2 be subspaces of V . Assume that $\{v_1, v_2, \dots, v_m\}$ is a set of generators (a generating set) of V_1 and $\{u_1, u_2, \dots, u_n\}$ is a set of generators of V_2 . Prove that $\{v_1, \dots, v_m, u_1, u_2, \dots, u_n\}$ is also a set of generators of $V_1 + V_2$.

Exercise 56. Prove that $V = V_1 \oplus V_2$ ¹ if and only if each $v \in V$ has a unique representation in the form $v = v_1 + v_2$, where $v_1 \in V_1$ and $v_2 \in V_2$.

3.2. Dimension and coordinates

Exercise 57. a) Write $v = (3, 0, -6)$ as a linear combination of u_1, u_2 and u_3 if possible, where $u_1 = (1, -1, 2)$, $u_2 = (2, 4, -2)$, $u_3 = (1, 2, -4)$.

b) Express the polynomial $v = t^2 + 4t - 3$ over $\mathbb{R}$ as a linear combination of the polynomials $p_1 = t^2 - 2t + 5$, $p_2 = 2t^2 - 3t$, $p_3 = t + 3$.

c) Find a condition on a, b, c so that $w = (a, b, c)$ is a linear combination of $u = (1, -3, 2)$ and $v = (2, -1, 1)$.

d) Is the vector $(3, -1, 0, -1)$ spanned by the vectors $(2, -1, 3, 2)$, $(-1, 1, 1, -3)$ and $(1, 1, 9, -5)$?

Exercise 58. Determine whether the following vectors are linearly dependent or linearly independent:

a) $v_1 = (4; -2; 6), v_2 = (-6; 3; -9)$.

b) $v_1 = (2; 3; -1), v_2 = (3; -1; 5), v_3 = (-1; 3; -4)$.

c) $v_1 = (1; 2; 3), v_2 = (3; 6; 7), v_3 = (-3; 1; 3), v_4 = (0; 4; 2)$.

d) $u_1 = 1 + 2x, u_2 = 3x - x^2, u_3 = 2 - x + x^2$.

Exercise 59. Let V be the vector space of functions from $\mathbb{R}$ to $\mathbb{R}$. Show that $f, g, h \in V$ are linearly independent, where $f(t) = \sin t$, $g(t) = \cos t$, $h(t) = t$.

Exercise 60. Let v_1, v_2 and v_3 be linearly independent vectors in a vector space V .

a) Prove that $\{v_1 - v_2, v_2 - v_3, v_1 + v_2 + v_3\}$ is linearly independent.

b) Prove that $\{v_1 - v_2, v_2 - v_3, v_1 - 2v_2 + v_3\}$ is linearly dependent.

c) For which values of a is the set $\{v_1 - v_2, v_2 - v_3, v_1 + av_2 + v_3\}$ linearly independent?

Exercise 61. Determine whether the set S is a basis of $\mathbb{R}^3$.

¹We say that V is a direct sum of V_1 and V_2 and write $V = V_1 \oplus V_2$ if $V_1 + V_2 = V, V_1 \cap V_2 = \{0\}$.

- a) $S = \{(1, 2, 1), (1, 1, 1)\}$.
 b) $S = \{(1, 1, 7), (3, 1, -3), (2, 1, 2)\}$.
 c) $S = \{(1, 0, -1), (1, 2, 1), (0, -3, 2)\}$.

Exercise 62. Determine whether the set S in $P_2[x]$ is a basis.

- a) $S = \{1 + x, 2 + x + x^2, 3 - 2x + x^2\}$.
 b) $S = \{x^2 + 3x - 2, 2x^2 + 5x - 3, -x^2 - 4x + 3\}$.

Exercise 63. Find a basis and the dimension of subspace W of $\mathbb{R}^3$ in the following cases:

- a) $W = \{(a, a + b, a - 2b) \mid a, b \in \mathbb{R}\}$; b) $W = \{(x, y, z) \mid x + y + z = 0\}$;
 c) $W = \{(x, y, z) \mid x - 2y + z = 0 \text{ and } 2x - 3y + z = 0\}$.

Exercise 64. Let W be a subspace of $\mathbb{R}^4$ spanned by the vectors

$$u_1 = (1, -2, 5, -3), u_2 = (2, 3, 1, -4), u_3 = (3, 8, -3, -5).$$

- a) Find a basis and the dimension of W .
 b) Extend the basis of W found in part a) to a basis of the whole space $\mathbb{R}^4$.

Exercise 65. Find the coordinate of u relative to (with respect to) the basis S in the following cases:

- a) $S = \{(1, 1), (0, -2)\}$, $u = (2, -1)$ in $\mathbb{R}^2$.
 b) $S = \{(1, 0, 0), (0, 1, 0), (1, 1, 1)\}$, $u = (4, -2, 9)$ in $\mathbb{R}^3$.
 c) $S = \{(1, 2, 3), (1, 2, 0), (0, -6, 2)\}$, $u = (3, -3, 0)$ in $\mathbb{R}^3$.
 d) $S = \{1 + x, 2 + x + x^2, 3 - 2x + x^2\}$, $u = 6 - 7x + x^2$ in $P_2[x]$.

Exercise 66. Find the coordinate of x relative to the basis S' , where

$$S = \{(1, 1), (1, -1)\}, S' = \{(0, 1), (1, 2)\}, [x]_S = \begin{bmatrix} 3 \\ -3 \end{bmatrix}.$$

Exercise 67. Consider the subspaces $U = \text{span}\{u_1, u_2, u_3\}$ and $W = \text{span}\{w_1, w_2, w_3\}$ of $\mathbb{R}^3$, where $u_1 = (1, 1, -1)$, $u_2 = (2, 3, -1)$, $u_3 = (3, 1, -5)$ and $w_1 = (1, -1, -3)$, $w_2 = (3, -2, -8)$, $w_3 = (2, 1, -3)$. Show that $U = W$.

Exercise 68. Let $v_1 = 1, v_2 = 1 + x, v_3 = x + x^2, v_4 = x^2 + x^3$ be vectors in $P_3[x]$.

- a) Prove that $B = \{v_1, v_2, v_3, v_4\}$ is a basis of $P_3[x]$.
 b) Find the coordinate of $v = 2 + 3x - x^2 + 2x^3$ with respect to B .

Exercise 69. Let E be the standard basis of $P_3[x]$ and $B = \{1, 1 + x, (1 + x)^2, (1 + x)^3\}$.

- a) Prove that B is a basis of $P_3[x]$.
- b) Find the change-of-basis matrices from E to B , and from B to E .
- c) Find the coordinate of $v = 2 + 2x - x^2 + 3x^3$ with respect to the basis B .

3.3. Rank

Exercise 70. Find the rank of the following vector system in $P_3[x]$:

$$v_1 = 1 + x^2 + x^3, v_2 = x - x^2 + 2x^3, v_3 = 2 + x + 3x^3, v_4 = -1 + x - x^2 + 2x^3.$$

Exercise 71. Find the rank of the following matrices:

$$\text{a) } A = \begin{bmatrix} 1 & 3 & 5 & -1 \\ 2 & -1 & -1 & 4 \\ 5 & 1 & -1 & 7 \\ 7 & 7 & 9 & 1 \end{bmatrix} \quad \text{b) } B = \begin{bmatrix} 4 & 3 & -5 & 2 & 3 \\ 8 & 6 & -7 & 4 & 2 \\ 4 & 3 & -8 & 2 & 7 \\ 4 & 3 & 1 & 2 & -5 \\ 8 & 6 & -1 & 4 & -6 \end{bmatrix}.$$

Exercise 72. Find m such that the rank of $A = \begin{bmatrix} 1 & -1 & 1 & 2 \\ -1 & 2 & 2 & 1 \\ 1 & 0 & 4 & m \end{bmatrix}$ equal to 2.

3.4. Linear systems of equations revisited

Exercise 73. Let us consider the following system:
$$\begin{cases} x_1 + 2x_2 - x_3 + mx_4 & = 4 \\ -x_1 - x_2 + 3x_3 + 2x_4 & = k \\ 2x_1 - x_2 - 3x_3 + (m-1)x_4 & = 3 \\ x_1 + x_2 + x_3 + 2mx_4 & = 5. \end{cases}$$

- a) Solve the system for $m = 2, k = 5$.
- b) Find m and k such that the system has a unique solution.
- c) Find m and k such that the system has infinitely many solutions. Then, compute the solutions.

Exercise 74. Solve the following system of linear equations depending on a parameter m :

$$\begin{cases} x_1 - x_2 + mx_3 - 2x_4 = 1 \\ 2x_1 - x_2 + 2x_3 + 3x_4 = -1 \\ -3x_1 + 3x_2 + (2m-8)x_3 - 12x_4 = 5 \\ -7x_1 + 3x_2 + (m-8)x_3 - 14x_4 = 5. \end{cases}$$

Exercise 75. Find the dimension and a basis of the solution space of the following system:

a) $2x_1 - 5x_2 + 3x_3 = 0.$

c)
$$\begin{cases} 2x_1 - x_2 + 3x_3 - 2x_4 + 4x_5 = 0 \\ 4x_1 - 2x_2 + 5x_3 + x_4 + 7x_5 = 0 \\ 2x_1 - x_2 + x_3 + 8x_4 + 2x_5 = 0. \end{cases}$$

b)
$$\begin{cases} x + 2y + z - 3t = 0 \\ 2x + 4y + 4z - t = 0 \\ 3x + 6y + 7z + t = 0. \end{cases}$$

d)
$$\begin{cases} x_1 - x_2 + 2x_3 + 2x_4 - x_5 &= 0 \\ x_1 - 2x_2 + 3x_3 - x_4 + 5x_5 &= 0 \\ 2x_1 + x_2 + x_3 + x_4 + 3x_5 &= 0 \\ 3x_1 - x_2 - 2x_3 - x_4 + x_5 &= 0. \end{cases}$$

Chapter 4

Linear mappings and transformations

4.1-4.3. Linear mappings

Exercise 76. If $\alpha_1 = (1, -1), \alpha_2 = (2, -1), \alpha_3 = (-3, 2), \beta_1 = (1, 0), \beta_2 = (0, 1), \beta_3 = (1, 1)$, is there a linear transformation T from $\mathbb{R}^2$ to $\mathbb{R}^2$ such that $T(\alpha_i) = \beta_i$ for $i = 1, 2, 3$?

Exercise 77. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear mapping for which $T(1, 2) = (2, 3)$ and $T(0, 1) = (1, 4)$. Find a formula for T , that is, find $T(a, b)$ for arbitrary a and b .

Exercise 78. Suppose $a, b \in \mathbb{R}$. Define $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by $T(x, y, z) = (2x - 4y + 3z + a, 6x + bxyz)$. Show that T is linear if and only if $a = b = 0$.

Exercise 79. Let T be the mapping from $\mathbb{R}^3$ to $\mathbb{R}^3$ defined by

$$T(x_1, x_2, x_3) = (x_1 - x_2 + 2x_3, 2x_1 + x_2, -x_1 - 2x_2 + 2x_3).$$

a) Verify that T is a linear transformation.

b) Show that $(a, b, c) \in \text{Im}T$ if and only if $-a + b + c = 0$.

c) Find a basis of $\text{Im}T$.

d) Find a basis of $\text{Ker} T$.

Exercise 80. Find a basis for $\text{Ker}(T)$ and $\text{Im}(T)$, where

$$T: \mathbb{R}^4 \rightarrow \mathbb{R}^3, T(x, y, z, w) = (4x - 5y + 5z + 2w, -2x + 2y - w, -y + 5z).$$

Exercise 81. Let $T: V \rightarrow U$ be linear, and suppose $v_1, \dots, v_n \in V$ have the property that their images $T(v_1), \dots, T(v_n)$ are linearly independent. Show that the vectors $v_1, \dots, v_n$ are also linearly independent.

Exercise 82. Suppose $T: V \rightarrow U$ be an injective linear map and $v_1, \dots, v_m$ are linearly independent in V . Show that $T(v_1), \dots, T(v_m)$ are linearly independent in U .

Exercise 83. Give an example of a linear map $T: \mathbb{R}^4 \rightarrow \mathbb{R}^4$ such that $\text{Im}T = \text{Ker}T$.

Exercise 84. Prove that there does not exist a linear map $T: \mathbb{R}^5 \rightarrow \mathbb{R}^5$ such that $\text{Im}T = \text{Ker}T$.

Exercise 85. Suppose that $T: \mathbb{R}^4 \rightarrow \mathbb{R}^2$ is a linear map fulfilling

$$\text{Ker}T = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 \mid x_1 = 5x_2 \text{ and } x_3 = 7x_4\}.$$

Prove that T is surjective.

Exercise 86. Prove that there does not exist a linear map $T: \mathbb{R}^5 \rightarrow \mathbb{R}^2$ such that

$$\text{Ker}T = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 \mid x_1 = 3x_2 \text{ and } x_3 = x_4 = x_5\}.$$

Exercise 87. Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear map defined by $f(x_1, x_2, x_3) = (3x_1 + x_2 - x_3, 2x_1 + x_3)$. Find the matrix of f with respect to the standard basis.

Exercise 88. Find the matrix of T with respect to the bases $\mathcal{B}$ and $\mathcal{B}'$, where

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3, T(x, y) = (-x, y, x + y), \mathcal{B} = \{(1, 1), (1, -1)\}, \mathcal{B}' = \{(0, 1, 0), (0, 0, 1), (1, 0, 0)\}.$$

Exercise 89. Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a mapping defined by

$$f(x_1, x_2, x_3) = (x_1 + x_2 - x_3, x_1 - x_2 + x_3, -x_1 + x_2 + x_3).$$

Find the matrix of f with respect to the basis $B = \{v_1 = (1, 0, 0), v_2 = (1, 1, 0), v_3 = (1, 1, 1)\}$.

Exercise 90. Consider the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (2x, 4x - y, 2x + 2y - z)$. Show that T is invertible. Find formulas for T^{-1} , T^2 and T^{-2} .

Exercise 91. Let $f: P_2[x] \rightarrow P_4[x]$ be a map defined by $f(p) = p + x^2p, \forall p \in P_2[x]$.

- Prove that f is a linear map.
- Find the matrix of f with respect to the bases $E_1 = \{1, x, x^2\}$ of $P_2[x]$ and $E_2 = \{1, x, x^2, x^3, x^4\}$ of $P_4[x]$.
- Find the matrix of f with respect to the bases $E'_1 = \{1 + x, 2x, 1 + x^2\}$ of $P_2[x]$ and $E_2 = \{1, x, x^2, x^3, x^4\}$ of $P_4[x]$.

Exercise 92. Suppose $A = \begin{bmatrix} 1 & 3 & -1 \\ 2 & 0 & 5 \\ 6 & -2 & 4 \end{bmatrix}$ is the matrix of the linear transformation $f: P_2[x] \rightarrow P_2[x]$ with respect to the basis $B = \{v_1, v_2, v_3\}$, where

$$v_1 = 3x + 3x^2, v_2 = -1 + 3x + 2x^2, v_3 = 3 + 7x + 2x^2.$$

a) Find $f(v_1), f(v_2), f(v_3)$.

b) Find $f(1 + x^2)$.

Exercise 93. Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Prove that

$$\text{rank}(AB) \leq \min \{ \text{rank } A, \text{rank } B \}.$$

Exercise 94. Let A and B be matrices of size $m \times n$. Prove that $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$.

Chapter 5

Eigenvalues and eigenvectors

5.1. Eigenvalues and eigenvectors

Exercise 95. Find the eigenvalues and a basis for each eigenspace of the following matrices:

a) $\begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 1 \\ -1 & -2 & 4 \end{bmatrix}$

c) $\begin{bmatrix} 0 & 1 & 0 \\ -4 & 4 & 0 \\ -2 & 1 & 2 \end{bmatrix}$

d) $\begin{bmatrix} 4 & -5 & 2 \\ 5 & -7 & 3 \\ 6 & -9 & 4 \end{bmatrix}$

Exercise 96. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3, T(x, y, z) = (-2x + 2y - 3z, 2x + y - 6z, -x - 2y)$. Find all eigenvalues and a basis for each eigenspace of T .

Exercise 97. Let $f: P_2[x] \rightarrow P_2[x]$ be a linear transformation defined by

$$f(a_0 + a_1x + a_2x^2) = (5a_0 + 6a_1 + 2a_2) - (a_1 + 8a_2)x + (a_0 - 2a_2)x^2.$$

Find the eigenvalues and eigenvectors of f .

5.2-5.3. Properties of eigenvalues and eigenvectors, diagonalization

Exercise 98. Suppose that $T: V \rightarrow V$ is linear with $\text{rank } T = k$. Prove that T has at most $k + 1$ distinct eigenvalues.

Exercise 99. Suppose that $T: V \rightarrow V$ is linear and there exist nonzero vectors v and w in V such that $T(v) = 3w$ and $T(w) = 3v$. Prove that either 3 or -3 is an eigenvalue of T .

Exercise 100. Suppose that $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is linear and that $-4, 5, \sqrt{6}$ are eigenvalues of T . Prove that there exists $x \in \mathbb{R}^3$ such that $T(x) - 7x = (-4, 5, \sqrt{6})$.

Exercise 101. Let $\lambda_1, \dots, \lambda_n$ be a list of distinct real numbers. Prove that the list $e^{\lambda_1 x}, \dots, e^{\lambda_n x}$ is linearly independent in the vector space of real-valued functions on $\mathbb{R}$. (Hint: Let $V = \text{span}\{e^{\lambda_1 x}, \dots, e^{\lambda_n x}\}$ and define $T: V \rightarrow V$ by $T(f) = f'$.)

Exercise 102. Diagonalize the following matrices (if possible):

$$\begin{array}{lll} \text{a) } A = \begin{bmatrix} -14 & 12 \\ -20 & 17 \end{bmatrix} & \text{c) } C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} & \text{e) } E = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix} \\ \text{b) } B = \begin{bmatrix} 1 & 0 \\ 6 & -1 \end{bmatrix} & \text{d) } D = \begin{bmatrix} 2 & 1 & -2 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix} & \end{array}$$

Exercise 103. Suppose that $A, B \in M_3(\mathbb{R})$ have 2, 6, 7 as eigenvalues. Prove that there exists an invertible matrix $P \in M_3(\mathbb{R})$ such that $B = P^{-1}AP$.

Exercise 104. Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation defined by

$$f(x_1, x_2, x_3) = (2x_1 - x_2 - x_3, x_1 - x_2 + x_3, -x_1 + x_2 + 2x_3).$$

Diagonalize the transformation f .

Exercise 105. Find a basis of $\mathbb{R}^3$ such that the matrix of $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with respect to this basis is a diagonal matrix, where

$$f(x_1, x_2, x_3) = (2x_1 + x_2 + x_3, x_1 + 2x_2 + x_3, x_1 + x_2 + 2x_3).$$

Exercise 106. Let $T: P_2[x] \rightarrow P_2[x]$ be the linear transformation given by

$$T(a + bx + cx^2) = (a + 3b + 3c) + (3a + b + 3c)x + (3a + 3b + c)x^2.$$

Find a basis B of $P_2[x]$ (if possible) such that the matrix of T with respect to B is diagonal. (Diagonalize the transformation T .)

Exercise 107. The trace of an $n \times n$ matrix A is defined to be the sum of the elements on the main diagonal, i.e., $\text{tr}(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}$. Prove that

a) The trace $\text{tr}: M_n(\mathbb{R}) \rightarrow \mathbb{R}$ is a linear mapping, that is,

$$\text{i) } \text{tr}(A + B) = \text{tr}(A) + \text{tr}(B), \quad \text{ii) } \text{tr}(cA) = c \text{tr}(A);$$

$$\text{b) } \text{tr}(A) = \text{tr}(A^T), \quad \text{c) } \text{tr}(AB) = \text{tr}(BA), \quad \text{d) } \text{tr}(P^{-1}AP) = \text{tr } A.$$

Exercise 108. Let $A \in M_n(\mathbb{C})$ be an invertible matrix and $0 \neq \lambda \in \mathbb{C}$. Show that λ is an eigenvalue of A if and only if $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} .

Exercise 109. Let $p(x) \in \mathbb{C}[x]$ be a polynomial and let A be a square matrix. Show that if λ is an eigenvalue of A then $p(\lambda)$ is an eigenvalue $p(A)$.

Exercise 110. Let $A \in M_n(\mathbb{C})$ and let $p \in \mathbb{C}[x]$ be a polynomial such that $p(A) = 0$. Prove that any eigenvalue λ of A satisfies $p(\lambda) = 0$.

Chapter 6

Euclidean spaces, orthogonality

6.1. Inner products

Exercise 111. Verify that the following is an inner product on $\mathbb{R}^2$ where $u = (x_1, x_2)$ and $v = (y_1, y_2)$:

$$\langle u, v \rangle = x_1 y_1 - 2x_1 y_2 - 2x_2 y_1 + 5x_2 y_2.$$

Exercise 112. Find the values of k so that the following is an inner product on $\mathbb{R}^2$ where $u = (x_1, x_2)$ and $v = (y_1, y_2)$:

$$\langle u, v \rangle = x_1 y_1 - 3x_1 y_2 - 3x_2 y_1 + kx_2 y_2.$$

Exercise 113. Determine whether each of the following defines an inner product on $P_3[x]$:

- a) $\langle p, q \rangle = p(0)q(0) + p(1)q(1) + p(2)q(2)$
- b) $\langle p, q \rangle = p(0)q(0) + p(1)q(1) + p(2)q(2) + p(3)q(3)$
- c) $\langle p, q \rangle = \int_{-1}^1 p(x)q(x)dx.$

If yes, compute $\langle p, q \rangle$, where $p = 2 - 3x + 5x^2 - x^3$, $q = 4 + x - 3x^2 + 2x^3$.

Exercise 114. Suppose that $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear transformation such that $\|Tv\| \leq \|v\|$ for any $v \in V$. Prove that $T - \sqrt{2}I$ is invertible.

Exercise 115. (a) Prove that for any $u, v, w \in \mathbb{R}^n$.

$$\left\| w - \frac{1}{2}(u + v) \right\|^2 = \frac{\|w - u\|^2 + \|w - v\|^2}{2} - \frac{\|u - v\|^2}{4}.$$

- (b) Suppose that A is a subset of $\mathbb{R}^n$ such that if $u, v \in A$, then $\frac{1}{2}(u + v) \in A$. Let $w \in \mathbb{R}^n$. Show that there is at most one vector $u \in A$ satisfying

$$\|w - u\| \leq \|w - v\| \quad \text{for all } v \in C.$$

6.2. Orthogonality

Exercise 116. Let the inner product on $P_2[x]$ be defined as $\langle p, q \rangle = \int_{-1}^1 p(x)q(x)dx$, where $p, q \in P_2[x]$.

- Apply the Gram-Schmidt process to the basis $\{1, x, x^2\}$ to get an orthonormal basis $\mathcal{A}$.
- Find the coordinate vector $[r]_{\mathcal{A}}$, where $r = 2 - 3x + 3x^2$.

In the following exercises (Ex 117-Ex 121), we consider $\mathbb{R}^n$ or $M_{n \times 1}(\mathbb{R})$ with the standard inner product.

Exercise 117. Find vectors $u, v \in \mathbb{R}^2$ such that u is a scalar multiple of $(1, 3)$, v is orthogonal to $(1, 3)$ and $(1, 2) = u + v$.

Exercise 118. Let consider the following vector system S in $\mathbb{R}^4$:

$$S = \{u_1 = (1, 1, 1, 1), u_2 = (1, 1, -1, -1), u_3 = (1, -1, 1, -1), u_4 = (1, -1, -1, 1)\}.$$

- Show that S is orthogonal and a basis of $\mathbb{R}^4$.
- Write $v = (1, 3, -5, 6)$ as a linear combination of u_1, u_2, u_3, u_4 .
- Find the coordinates of an arbitrary vector $v = (a, b, c, d)$ in $\mathbb{R}^4$ relative to the basis S .
- Normalize S to obtain an orthonormal basis of $\mathbb{R}^4$.

Exercise 119. Use the Gram-Schmidt process to transform the basis B into an orthonormal basis.

- $B = \{(1, 1), (0, 1)\}$.
- $B = \{(1, -2, 2), (2, 2, 1), (2, -1, -2)\}$,
- $B = \{(1, 2, -2), (0, 1, -2), (-1, 3, 11)\}$,
- $B = \{(3, 4, 0, 0), (-1, 1, 0, 0), (2, 1, 0, -1), (0, 1, 1, 0)\}$.

Exercise 120. Let $v_1 = (1, 1, 0, 0, 0)$, $v_2 = (0, 1, -1, 2, 1)$, $v_3 = (2, 3, -1, 2, 1)$, and

$$V = \{x \in \mathbb{R}^5 \mid x \perp v_i, i = 1, 2, 3\}.$$

- Prove that V is a subspace of $\mathbb{R}^5$.
- Find a basis and the dimension of V .

Exercise 121. Let W be a the solution space of the following homogeneous system:

$$\begin{cases} x + y - z + w &= 0, \\ 2x + y + z + 2w &= 0. \end{cases}$$

- Find an orthonormal basis for W .

(b) Find an orthonormal basis for $W^\perp$.

(c) Find a system of linear equations for which $W^\perp$ is its solution space.

Exercise 122. Find the (orthogonal) projection of $u = (1, 3, -2, 4)$ on $v = (2, -2, 4, 5)$.

Exercise 123. Let $v_1 = (2, 2, 1)$, $v_2 = (2, 5, 4)$. Find the (orthogonal) projection of $v = (3, -2, 1)$ onto $U = \text{span}\{v_1, v_2\}$.

6.3. Least square approximations

Exercise 124. In $\mathbb{R}^4$, let $U = \text{span}\{(1, 1, 0, 0), (1, 1, 1, 2), (2, 2, 1, 2)\}$. Find $u \in U$ such that $\|u - (1, 2, 3, 4)\|$ is as small as possible.

Exercise 125. Find $a, b \in \mathbb{R}$ such that

$$(a + b - 1)^2 + (a + b - 2)^2 + (b - 3)^2 + (2b - 4)^2$$

is as small as possible.

Exercise 126. Let $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$. Find a column vector $\tilde{X} \in M_{3 \times 1}(\mathbb{R})$ for which minimizes the function $f(X) = \|AX - B\|$ defined for all $X \in M_{3 \times 1}(\mathbb{R})$.

Exercise 127. Find the least square solution of the following system:

$$\begin{bmatrix} 1 & 2 & -2 \\ 1 & 0 & -2 \\ 2 & 8 & -4 \end{bmatrix} X = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} \quad \text{for } X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Then, explain the meaning of the least square solutions.

6.4. Orthogonal diagonalization

Exercise 128. Orthogonally diagonalize of the following symmetric matrices:

$$\begin{array}{lll} \text{a) } \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} & \text{b) } \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} & \text{c) } \begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & 2 \\ 0 & 2 & 5 \end{bmatrix}. \end{array}$$

6.5. Quadratic forms

Exercise 129. Determine the definiteness of the following quadratic form on $\mathbb{R}^3$:

a) $\omega_1(x_1, x_2, x_3) = x_1^2 + 5x_2^2 - 4x_3^2 + 2x_1x_2 - 4x_1x_3.$

b) $\omega_2(x_1, x_2, x_3) = 2x_1^2 + x_2^2 + 3x_3^2 + 2x_1x_2 + 2x_1x_3.$

c) $\omega_3(x_1, x_2, x_3) = x_1x_2 + 4x_1x_3 + x_2x_3.$

Exercise 130. Find the values of a such that the following quadratic forms are positive definite:

- a) $5x_1^2 + x_2^2 + ax_3^2 + 4x_1x_2 - 2x_1x_3 - 2x_2x_3$. c) $x_1^2 + x_2^2 + 5x_3^2 + 2ax_1x_2 - 2x_1x_3 + 4x_2x_3$.
 b) $2x_1^2 + x_2^2 + 3x_3^2 + 2ax_1x_2 + 2x_1x_3$.

Exercise 131. Orthogonally diagonalize the following quadratic forms:

- a) $x_1^2 + x_2^2 + x_3^2 + 2x_1x_2$. c) $7x_1^2 + 6x_2^2 + 5x_3^2 - 4x_1x_2 + 4x_2x_3$.
 b) $7x_1^2 - 7x_2^2 + 48x_1x_2$.

Exercise 132. Transform the following quadric surface to the principal axes:

$$2x^2 + 6y^2 + 14z^2 - 6xy + 2xz + 6yz + 2x - y + z = 0.$$

Exercise 133. Classify the following quadratic curves:

- a) $2x^2 - 4xy - y^2 + 8 = 0$. c) $2x^2 + 4xy + 5y^2 = 24$.
 b) $x^2 + 2xy + y^2 + 8x + y = 0$.

Exercise 134. Classify the following quadric surfaces:

- a) $x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 = 4$. b) $2x_1^2 + 2x_2^2 + 3x_3^2 - 2x_1x_2 - 2x_2x_3 = 16$.
 c) $2xy + 2yz + 2xz - 6x - 6y - 4z = 0$.

Exercise 135. Let $Q(x_1, x_2, x_3) = 9x_1^2 + 7x_2^2 + 11x_3^2 - 8x_1x_2 + 8x_1x_3$.

Find $\max_{x_1^2+x_2^2+x_3^2=16} Q(x_1, x_2, x_3)$ and $\min_{x_1^2+x_2^2+x_3^2=16} Q(x_1, x_2, x_3)$.

Exercise 136. Let $Q(x, y, z) = 9x^2 + 7y^2 + 11z^2 - 8xy + 8xy$ with $x, y, z \in \mathbb{R}$.

Find $\max_{Q(x,y,z)=16} x^2 + y^2 + z^2$ and $\min_{Q(x,y,z)=16} x^2 + y^2 + z^2$.

Exercise 137. Is there an orthogonal matrix $A \in M_3(\mathbb{R})$ such that

$$A \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad A \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ 0 \end{bmatrix}?$$

Exercise 138. Is there a symmetric matrix $A \in M_3(\mathbb{R})$ such that

$$A \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad A \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}?$$